# Exploratory Analysis of Online Payments Fraud Detection using Machine Learning Project Documentation

## 1. Introduction

**Project Title:** Online Payments Fraud Detection using Machine Learning

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## 2. Project Overview

To detect fraudulent online payment transactions using Machine Learning techniques and reduce financial losses for customers and financial institutions.

**Features:**

- Real-time fraud detection
- Transaction data preprocessing
- Fraud probability scoring

## 3. Architecture

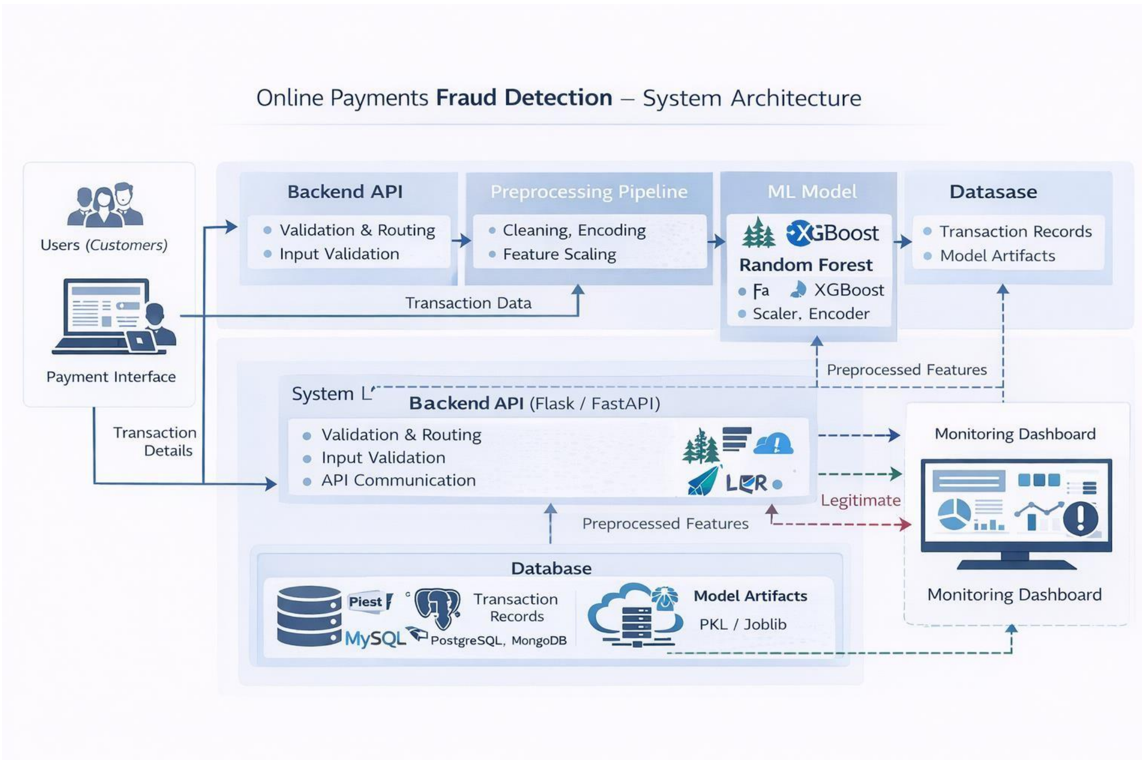
**Frontend:**
HTML, CSS, JavaScript dashboard for monitoring transactions and displaying fraud results.

**Backend:**
Flask/FastAPI-based Python server handling transaction requests, validation, and prediction logic.
**Model Layer:**

Trained Machine Learning model (Random Forest / XGBoost) with scaler and encoder artifacts.

**Database Layer:**
Stores transaction records, fraud logs, and model artifacts.

## System Architecture Diagram




# 4. Setup Instructions

## Prerequisites:

- Python 3.x
- Flask or FastAPI
- Pandas
- NumPy
- Scikit-learn
- Joblib

- MySQL/MongoDB (optional)

### Installation:

Install dependencies using pip and run the backend application.

## 5. Folder Structure

templates/ → HTML dashboard files static/

→ CSS, JS, Images main.py → Flask/FastAPI

backend model_training.ipynb → ML model training

notebook fraud_model.pkl → Trained model

scaler.pkl → Saved scaler encoder.pkl →

Saved encoder

## 6. Running the Application python main.py

http://localhost:5000

## 7. API Documentation

Accepts transaction details and returns fraud prediction.

## 8. Authentication

Not implemented in current version (Future scope: JWT-based authentication and secure token validation).

## 9. User Interface The

UI provides:

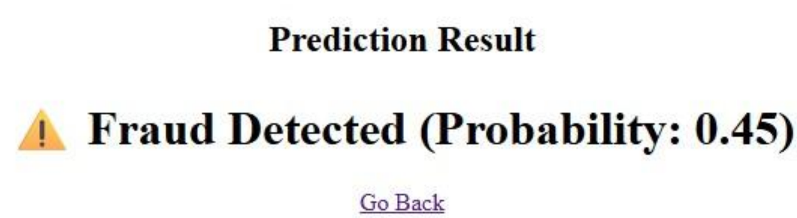
- Transaction input form (for testing)
- Fraud detection result display

- Fraud monitoring dashboard
- Statistical charts for fraud trends
- Alert notifications

## 10. Testing

Model evaluated using accuracy, confusion matrix, ROC-AUC.

## 11. Screenshots / Demo


## 12. Known Issues

- Performance depends on dataset quality
- Imbalanced data affects recall
- Requires periodic retraining to handle evolving fraud patterns

## 13. Future Enhancements

- Deep Learning-based fraud detection
- Real-time streaming using Kafka

- Cloud deployment (AWS/Azure/GCP)
- Mobile fraud alert system
- Cross-bank fraud intelligence sharing
- Advanced risk scoring system